Data Availability and Description of Public Datasets

This document provides a comprehensive description of the public datasets and standard test systems used in the study titled "**Decision-Dependent Distributionally Robust Scheduling of Price-Responsive EV Aggregators in Distribution Networks.**"

To ensure reproducibility, the processed inputs, including charging sessions, price-response data, wind errors, and market prices, are publicly available at our GitHub repository [8]: https://github.com/dishiminthu/DD-DR-Scheduling-of-Price-Responsive-EVAs

The original data sources and their respective licenses are documented within the repository. Below is a detailed breakdown of the data utilized:

1. Electric Vehicle (EV) Charging Sessions

a) Source: The UK Department for Transport's "Electric Chargepoint Analysis 2017: Domestics" dataset [1].
b) Scale & Processing: A total of 300,000 charging session records were used. To preserve the joint distribution of arrival time, plug-in duration, and energy demand, the sessions were bootstrapped to generate synthetic daily profiles with a Poisson-distributed number of sessions per day.
c) Fleet Composition: Based on data-driven user typologies [2], four distinct EV Aggregator (EVA) fleets were formed: two residential, one commercial, and one workplace fleet. The dataset is split into 300 training days, 65 validation days, and 200 independent test days.

2. Price-Response Dataset

a) Source: A public EV charging recommendation and price-response dataset [3].
b) Scale & Processing: The dataset contains 72,962 EV charging recommendations. Read with its documented labels, it encodes a threshold rule: plugged-in vehicles are recommended Vehicle-to-Grid (V2G) discharge when the price ratio exceeds one, and grid-to-vehicle (G2V) charging otherwise.
c) Usage: This data is used to inform the direction and saturating shape of the price-response map (specifically, the jump in V2G share at the reference price). Because the data records recommendations rather than observed choices, the exact strength of the response for a heterogeneous fleet is treated as a stated scenario in the optimization model.

3. Renewable Generation and Exogenous Uncertainty Data

a) Wind Errors: Derived from the day-to-day variability of historical wind speed data using the National Renewable Energy Laboratory (NREL) Wind Integration National Dataset (WIND) Toolkit [4].
b) Photovoltaic (PV) Errors: Derived from public 15-minute prosumer data.
c) Processing: Both PV and wind errors are scaled by a factor of 2.4 to represent stressed conditions and are truncated to the physical output range of the generation units.

4. Market Price Data

a) Source: California Independent System Operator (CAISO) Open Access Same-time Information System (OASIS) [5].
b) Usage: Energy and reserve capacity prices are extracted from a representative day of the California market. The peak-shaving capacity price is calculated as a function of the grid energy price and its daily average.

5. Standard Test Systems (Network Topology)

a) IEEE 33-Bus Feeder: Used at its standard peak load of 3.715 MW and 2.30 Mvar (derived from Baran & Wu, 1989 [6]).
b) IEEE 123-Node Feeder: Used as a balanced single-phase equivalent with 125 buses and a peak load of 3.49 MW (derived from Kersting, 1991 [7]).

Note: Both network topologies, branch ratings, and generator/PV/wind locations are standard public test cases adapted for the specific EV and renewable integration scenarios in this study.

References

[1] UK Department for Transport, "Electric Chargepoint Analysis 2017: Domestics," 2018. [Online]. Available: https://www.gov.uk/government/statistics/electric-chargepoint-analysis-2017-domestics

[2] J. R. Helmus, M. H. Lees, and R. van den Hoed, "A data driven typology of electric vehicle user types and charging sessions," Transp. Res. Part C: Emerg. Technol., vol. 115, p. 102637, 2020.

[3] V. Veeramsetty et al., "Dataset for EV charging recommendations and price response," Mendeley Data / IEEE DataPort, 2026.

[4] C. Draxl, A. Clifton, B.-M. Hodge, and J. McCaa, "The Wind Integration National Dataset (WIND) Toolkit," Appl. Energy, vol. 151, pp. 355–366, 2015.

[5] California Independent System Operator, "CAISO OASIS: Open Access Same-time Information System," 2025. [Online]. Available: http://oasis.caiso.com/

[6] M. E. Baran and F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," IEEE Trans. Power Del., vol. 4, no. 2, pp. 1401–1407, 1989.

[7] W. H. Kersting, "Radial distribution test feeders," IEEE Trans. Power Syst., vol. 6, no. 3, pp. 975–985, 1991.

[8] S. Di, "Code and data for decision-dependent distributionally robust scheduling of EV aggregators," GitHub repository, 2026. [Online]. Available: https://github.com/dishiminthu/DD-DR-Scheduling-of-Price-Responsive-EVAs